

Lanthanides, and they're used in everything from car exhausts to phones, and DVDs to fluorescent lighting. In fact, pretty much any product you pick up either contains lanthanides, or lanthanides were used in its manufacturing process. In technology, lanthanides are especially important as they're used in batteries, in the production of glass, magnets, memory, and various other uses. Their name is a little misleading though, because they're not actually 'rare' like gold is, and there are a fair bit of identified reserves across the globe. In fact, Thulium and Lutetium are two of the rarest rare earth elements, and it is estimated that each of them have reserves at least 200 times greater than the amount of gold that can be found!

BLAST PIT

China has a little over one third of all identified reserves of lanthanides on earth. Until the 1980s China was nowhere to be found on the scene of rare earth metal production, but thanks to their huge reserves, they were able to capture the market in under 20 years. By 2000 they'd priced every other country out of the game, because they had far cheaper labour, and each mine yielded more than competitors across the globe. What's important to remember is that since 2001, which is when China entered the world trade organisation, China ramped up manufacturing like mad. This means that while China is the world's largest miner of lanthanides, it's also the world's biggest consumer of them.

In 2010, China was in control of 95% of lanthanide production, and we all know what monopolies lead to. They were able to set their own price, and as a result, raised prices to the point where other countries had to consider starting up their own mining operations. Even now geologists across the globe are engaged in trying to find lanthanide deposits, especially deposits that would be cheap to mine.

China mines about 132,000 metric tonnes (MT) a year, and the world-wide production is about 210,000 MT. That's well over 60% of the world's production, but most of it is used by

China itself. The US produces about 26,000 MT, Myanmar makes 22,000 MT, Australia makes 21,000 MT and India is a distant fifth at about 3,000 MT. Spots 6 to 10 in order of production are Russia, Madagascar, Thailand, Brazil and Vietnam, all together totalling 8,400 MT.

China has about 44 million MT of reserves, while Vietnam and Brazil are estimated to have 22 million MT each, Russia has 18 million MT and India has about 7 million MT. If China was to go on strike, it would be the countries that we mentioned that would have to up their mining game, and fast. However, it would send prices skyrocketing temporarily, which would make everything else more expensive as well.

RESIDUE PROCESSING

Another problem that would crop up would be what to do with all the e-waste that's generated across the

ing, and manufacturing was taken up by a group of countries across the globe, theft would also have to start managing the e-waste that the manufacturing process produces, and would also have to account for all the e-waste that would accumulate just a few years later.

POWER UP

Large scale factories need power, and most places where the labour would be cheap enough to rival what China offers now, already have power problems. If, for example, India, Vietnam and Brazil wanted to capture a piece of the manufacturing pie that China abandoned when it went on strike, they would need to seriously ramp up energy production in order to be able to do so. China is already being criticised for using coal, the most harmful form of energy production in terms of climate change, in order to meet its manufacturing needs.

Countries like Australia, who supply a lot of this coal to China are also facing criticism for their environmental impact.

This is a problem that will merely change the offender, not the offense itself. If China were to go on strike, any country that tried to fill the gap in manufacturing would also end up turning to coal, because while it might be the most polluting, it's also the cheapest, quickest and

most reliable way to ramp up power production. If it was India who took over the majority of manufacturing, we'd soon become the world's largest polluter, just out of necessity.

FORGET ABOUT FREEMAN!

The first thing that happens when prices rise is that the poorest consumers get priced out of the game. A majority of the population of the Earth that is not still connected to the internet, but might soon be able to afford to – also called The Next Billion – lives in India. Although we enjoy some of the cheapest mobile bandwidth access costs in the world, we're also still try-



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Workers at a rare earth production factory in Jiangxi province, Central China

globe. While agreements and pacts that have been signed make it officially illegal to export a country's e-waste to another, it still happens. The protections were put in place to prevent rich countries from treating poorer countries as their dumping grounds, forcing poorer people of the world to undertake the dangerous and toxic activity of recycling e-waste. China, as one of the largest producers of e-waste in the world, also recycles the most. Plus while direct e-waste transportation might not be legal, it's perfectly fine to sell salvaged metals and components to one another. This means that if China were to go miss-